

MIT ACADEMY OF ENGINEERING

COURSE CODE: CS104

MARCH 2025

FY BTECH SEMESTER - II BACKLOG 2022 PATTERN 2024 - 2025 EXAMINATION

DEPARTMENT OF F.Y.BTECH

MID SEMESTER EXAMINATION

Essentials of Data Science

TIME : 1 HOUR

MAX MARKS : 20

TOTAL NO OF QUESTIONS: 20

TOTAL NO OF PRINTED PAGES: 4

INSTRUCTIONS TO CANDIDATES:

1. Assume suitable data wherever necessary
2. Non programmable scientific calculators are allowed
3. Black figures to the right indicate full marks

- 1 Select correct option for the given question: [1] CO1 L2
- a = 9
 b = 3
 print(a % b)
 print(a / b)
- a) 0 b) 3 c) 0 d) Error
 3.0 3.0 3
- 2 What will be printed by the following code when it executes? [1] CO1 L3
- count = 0
 while True:
 print("MIT AOE Rocks!")
 count += 1
 if count == 3:
 break
- 3 What will be printed by the following code when it executes? [1] CO1 L3
- max_odd = None
 values = [2, 7, 9, 12, 15]
 for num in values:
 if num % 2 == 1:
 if max_odd is None or num > max_odd:
 max_odd = num
 print(max_odd)
- a) 7 b) 9 c) 15 d) 12

- 4 Find the alphabet character from following user defined function:
`def substring(s):
 return s[4:11]`
`print(substring('MIT Academy of Engineering'))`
a) 'Academy' b) 'Academ' c) 'cademy ' d) 't Aca'
- 5 What will happen if a recursive function does not have a base condition?
a) The function will execute successfully and return a value.
b) The function will call itself indefinitely, leading to a stack overflow
c) The function will terminate automatically after a few calls.
d) The function will return None.
- 6 Which of the following operators is used for exponentiation (raising a number to a power) in Python?
a) ** b) ^ c) ^^ d) *
- 7 What will be printed by the following code when it executes?
`i = 2
while i <= 4:
 j = 3
 while j >= 1:
 print(f"i={i}, j={j}, product={i * j}")
 j -= 1
 i += 1`
- 8 Which type of loop can be used to perform the following iteration: You choose a positive integer at random and then print the numbers from 1 up to and including the selected integer.
a) Only a for loop b) Only while loop
c) a for loop and while loop d) None
- 9 What will be printed by the following code when it executes?
`def outer_function(x):
 def inner_function(y):
 return x + y
 return inner_function(5)`
`print(outer_function(10))`
a) 10 b) 5 c) 15 d) Error

10	What will be the output of the following Python code?	[1]	CO1	L3
	<pre>def read_file(file_name): file_obj = open(file_name, "r") for line in file_obj: print(line.strip()) file_obj.close() read_file("sample.txt")</pre>			
	a) Prints only the first line of sample.txt b) Throws a syntax error c) Throws a file not found error always d) Prints each line of sample.txt without extra newlines			
11	Which of the following statements about Python tuples is TRUE?	[1]	CO2	L2
	a) Tuples are mutable b) Tuples allow duplicate values c) Tuples do not support indexing d) Tuples must contain only strings			
12	What will be printed by the following code when it executes?	[1]	CO2	L2
	<pre>s = {10, 20, 30} s.add(40) print(s)</pre>			
	a) {10, 20, 30} b) {10, 20, 30, 40} c) {10, 20, 40} d) Error			
13	What will be printed by the following code when it executes?	[1]	CO2	L2
	<pre>my_list = [10, 20, 30, 40] print(my_list.index(50))</pre>			
	a) 50 b) None c) ValueError d) KeyError			
14	From the given syntax will create an empty dictionary?	[1]	CO2	L1
	a) DICT() b) DICT{} c) DICT=() d) DICT={}			
15	What will be the output of the following code?	[1]	CO2	L2
	<pre>my_list = [1, 2, 3, 4] my_list.append([5, 6]) print(my_list)</pre>			

- a) [1, 2, 3, 4, [5, 6]] b) [1, 2, 3, 4, (5, 6)]
 c) TypeError d) [1, 2, 3, 4, 5, 6]
- 16** What will be printed by the following code when it executes? **[1]** **CO2** **L3**
`x = [[(i,j) for j in range(2)] for i in range(2)]`
`print(x)`
- 17** Which of the following is an example of an ordinal attribute? **[1]** **CO2** **L1**
 a) PRN of a student b) Temperature in Celsius
 c) Rank in a competition d) Weight of a person
- 18** What will be printed by the following code when it executes? **[1]** **CO2** **L2**
`students = {`
`101: {'name': 'Alice', 'grade': 'A', 'age': 20},`
`102: {'name': 'Bob', 'grade': 'B', 'age': 21},`
`103: {'name': 'Charlie', 'grade': 'A', 'age': 22}`
`}`

`# Perform the following operation:`
`students[104] = {'name': 'David', 'grade': 'C', 'age': 20}`
`print(students)`
 a) A KeyError will occur
 b) The dictionary remains unchanged
 c) A new entry {104: {'name': 'David', 'grade': 'C', 'age': 20}} is added
 d) The program will crash
- 19** Which of the following is the correct syntax for a lambda function that returns the square of a number? **[1]** **CO2** **L2**
 a) `def square(x): return x ** 2`
 b) `lambda x => x ** 2`
 c) `square = lambda x { x ** 2 }`
 d) `lambda x: x ** 2`
- 20** Which of the following is an example of structured data? **[1]** **CO2** **L2**
 a) Social media posts b) Handwritten notes
 c) Audio recordings d) Data in relational database